

9.1 Introduction

Developed in the late seventies and early eighties, MIDI isn't really an audio format—rather, it's a protocol that defines musical notes in a precise format that computers can understand. This lets musical instruments (especially synthesizers) and computers talk to each other easily. When a musical instrument is connected to a PC via a MIDI interface (an interface that conforms to the MIDI protocol—this could be a serial port or a USB connector), it sends it:

1. A number indicating which note was pressed
2. When the note was pressed and subsequently released, and
3. How hard it was pressed

MIDI *Sequencers* are programs that record the above data and can now play it back at any speed and with any effect you wish to apply. The data is sent to your PC's MIDI synthesizer (which combines all the MIDI information to finally generate a sound). You can see the MIDI synthesizer by right-clicking on the volume icon on the taskbar, choosing “Adjust Audio Properties” and selecting the “Audio” tab.



Viewing your default MIDI synthesizer. For Windows XP users, it's the Microsoft GS Wavetable SW Synth

Wavetable Synthesis

In ye olde darke ages of computing, MIDI music was made up of single, boring tones, much like the ancient black-and-white cell phones. As MIDI grew, the need to emulate different instruments like pianos, guitars and violins became greater, and synthesizers

The great thing about MIDI is its sheer compactness—a 10 KB file can easily hold a minute of MIDI audio, making MIDI sequences the ideal choice for game music and any other application that requires a lot of audio in very little space.